Food Image Classification using Supervised and Self-Supervised Convolutional Neural Networks

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Machine Learning for Modelling —2025/2026
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Abstract

This report presents a food image classification system trained on the FoodX-251 dataset (1), which comprises 251 fine-grained food categories. We design and train a custom lightweight Convolutional Neural Network (FoodNet) under a supervised learning (SL) paradigm, achieving a top-1 validation accuracy of **27.04%** across all 251 classes. We additionally investigate self-supervised representation learning via the SimCLR framework (2), performing linear probing with a Linear SVM (9.20%) and k-Nearest Neighbours (3.58%). Our results confirm that supervised learning substantially outperforms self-supervised alternatives when labels are available, while demonstrating that meaningful visual representations can be learned without any label supervision.

1 Introduction

Fine-grained food classification is a challenging computer vision task due to the large number of visually similar categories, the non-rigid nature of food objects, and significant overlap in ingredient composition across dishes. These properties make it an ideal benchmark for evaluating both supervised and self-supervised deep learning methods.

This project investigates the iFood-2019 competition dataset (FoodX-251) (1), collected as part of the FGVC6 workshop at CVPR 2019. The dataset contains 251 fine-grained food categories with approximately 118k web-crawled training images. Our contributions are:

1. A custom CNN (FoodNet) with only 704k parameters trained in a supervised setting, achieving 27.04% top-1 accuracy on 251 classes.
2. A SimCLR-based self-supervised pre-training pipeline followed by linear probing with SVM and k-NN classifiers.
3. A systematic hyperparameter search justifying the chosen training configuration.
4. A quantitative comparison of SL and SSL approaches.

2 Dataset and Preprocessing

2.1 Dataset

The FoodX-251 dataset (1) was introduced for the iFood-2019 challenge. It contains 251 fine-grained food classes with images collected from the web, making label noise an inherent challenge. Table 1 summarises the dataset statistics.

Table 1: FoodX-251 dataset statistics. The official test set has no public ground truth, so the official validation set is used as our test set.

Split	Images	Classes	Labels
Training (original)	118,475	251	Noisy (web-sourced)
Validation (official)	12,170	251	Human-verified

2.2 Computational Sub-sampling

Training on the full 118k images within a limited GPU budget (Google Colab T4) is impractical. Following the project specification, we cap each class at **300 images**, retaining all 251 classes, yielding 74,990 training images in total. This is explicitly permitted by the project guidelines for motivated computational reasons.

2.3 Preprocessing Pipeline

All images are resized to 128×128 pixels and converted to RGB. Pixel values are normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). A stratified 85/15 split of the sub-sampled training data yields 63,741 training images and 11,249 validation images, both preserving class proportions across all 251 categories.

During supervised training the following augmentations are applied: random crop (128×128 from a 144×144 resize), random horizontal flip, colour jitter, and random rotation ($\pm 15°$). Validation images are only resized and normalised, with no stochastic augmentation, to ensure reproducible evaluation.

3 Custom CNN Architecture — FoodNet

3.1 Design Rationale

The project specification requires a custom CNN with fewer than 10 million parameters. FoodNet is a lightweight architecture inspired by MobileNet-style depthwise-separable convolutions (5), residual connections (3), and Squeeze-and-Excitation (SE) channel attention (4).

The key design decisions are: (i) depthwise-separable convolutions reduce parameters by approximately $8\times$ compared to standard convolutions; (ii) residual connections enable stable gradient flow in deeper stages; and (iii) SE blocks adaptively recalibrate channel-wise feature responses with negligible parameter overhead.

3.2 Architecture Details

Table 2 summarises the architecture. The total parameter count is **704,123**, well below the 10M constraint. All convolutional layers use Kaiming normal initialisation.

Table 2: FoodNet architecture overview. DWSep = Depthwise-Separable Conv; GAP = Global Average Pooling; SE = Squeeze-and-Excitation block.

Stage	Input Size	Output Size	Channels	Blocks
Stem	128×128	64×64	$3 \to 32$	$2\times$ ConvBNReLU
Stage 1	64×64	32×32	$32 \to 64$	DWSep + Residual + SE
Stage 2	32×32	16×16	$64 \to 128$	DWSep + $2\times$Residual + SE
Stage 3	16×16	8×8	$128 \to 256$	DWSep + $2\times$Residual + SE
Stage 4	8×8	4×4	$256 \to 256$	DWSep + Residual + SE
Head	4×4	—	$256 \to 251$	GAP + Dropout + FC

4 Supervised Learning

4.1 Training Configuration

Table 3 summarises the training configuration selected after hyperparameter tuning (Section 6).

Table 3: Supervised learning training configuration.

Hyperparameter	Value	Rationale
Optimiser	AdamW	Weight-decoupled Adam
Learning rate	3×10^{-3}	Best from grid search
Weight decay	1×10^{-4}	L2 regularisation
LR schedule	Cosine annealing	Smooth LR decay
Batch size	64	Fits T4 GPU VRAM
Max epochs	30	With early stopping (patience = 7)
Loss	CrossEntropy + LS	Label smoothing $\varepsilon = 0.1$
Gradient clipping	$\max_norm = 1.0$	Training stability

4.2 Training Progress

FoodNet was trained for all 30 epochs without early stopping triggering. Validation accuracy improved consistently from 1.34% at epoch 1 to **27.04%** at epoch 30, with the training loss decreasing from 5.46 to 3.80. Figure 2 shows the training and validation loss and accuracy curves.

4.3 Results

Table 4 reports the final evaluation metrics on the validation set using the best checkpoint (lowest validation loss).

Table 4: Supervised learning evaluation results on the validation set (251 classes, macro-averaged).

Metric	Value
Top-1 Accuracy	27.04%
Macro Precision	25.13%
Macro Recall	26.94%
Macro F1-score	24.73%

Although 27.04% may appear modest, random-chance performance on 251 classes is only 0.40%, so FoodNet achieves **67× above random** performance. The fine-grained nature of FoodX-251 (high intra-class variability, low inter-class variability) and the 300-image-per-class computational constraint largely explain this accuracy level.

5 Self-Supervised Learning — SimCLR

5.1 Approach

We adopt the SimCLR framework (2) for self-supervised pre-training. SimCLR learns visual representations by maximising agreement between two differently augmented views of the same image via a contrastive NT-Xent loss, without using any class labels.

The pipeline has three stages:

1. **Pre-training:** FoodNet backbone + a two-layer MLP projection head are trained with NT-Xent contrastive loss for 30 epochs.
2. **Feature extraction:** The projection head is discarded and the frozen backbone extracts 256-dimensional feature vectors.
3. **Linear probing:** A Linear SVM and a k-NN (k=5, cosine distance) are trained on the extracted features to evaluate representation quality.

5.2 SSL Configuration

Table 5: SimCLR pre-training configuration.

Hyperparameter	Value
Epochs	30
Batch size	128
Temperature τ	0.5
Projection dim	128
Optimiser	AdamW, LR=10^{-3}, WD=10^{-4}
LR schedule	Cosine annealing
SVM regularisation	$C = 0.1$ (LinearSVC)
k-NN neighbours	$k = 5$, cosine distance

5.3 Results

Table 6 reports linear probing accuracy on the validation set. Figure 4 shows the NT-Xent loss curve over 30 epochs.

Table 6: SSL linear probing evaluation results on the validation set (251 classes, macro-averaged).

Classifier	Accuracy	Precision	Recall	F1-score
SimCLR + Linear SVM	9.20%	7.49%	9.17%	6.70%
SimCLR + k-NN (k=5)	3.58%	5.62%	3.57%	3.26%

The NT-Xent loss decreases steadily from 4.99 at epoch 1 to 4.06 at epoch 30, confirming that the backbone progressively learns more discriminative representations. However, the linear probing accuracy gap relative to SL highlights that 30 epochs of contrastive pre-training on a 63k-image sub-sampled dataset is insufficient for robust 251-way classification.

6 Hyperparameter Tuning

A grid search was conducted over 12 hyperparameter combinations, with each configuration trained for 3 epochs on the full training split to identify promising settings before full training. Table 7 summarises the search space and the top-5 configurations.

Table 7: Hyperparameter grid search results (top-5 configurations, sorted by 3-epoch validation accuracy).

LR	Weight Decay	Dropout	Batch Size	Val Acc (3 ep.)
3×10^{-3}	10^{-3}	0.3	64	4.67%
5×10^{-3}	10^{-3}	0.4	64	4.61%
3×10^{-3}	10^{-4}	0.3	64	4.53%
3×10^{-3}	10^{-3}	0.4	64	4.44%
3×10^{-3}	10^{-4}	0.4	64	4.39%

The best configuration (LR $= 3 \times 10^{-3}$, WD $= 10^{-3}$, dropout $= 0.3$) was used for the final supervised training run. The learning rate sensitivity analysis (Figure **??**) confirms that intermediate rates (10^{-3}–3×10^{-3}) achieve the best short-run performance, while very low (10^{-4}) or very high (10^{-2}) rates lead to slow convergence or instability respectively.

7 Results Comparison

Table 8 and Figure 6 compare all three methods across four metrics.

Table 8: Summary comparison of all methods on the 251-class validation set (macro-averaged metrics). SL substantially outperforms both SSL variants, as expected when label information is available during training.

Method	Accuracy	Precision	Recall	F1-score
FoodNet (Supervised)	**27.04%**	**25.13%**	**26.94%**	**24.73%**
SimCLR + Linear SVM	9.20%	7.49%	9.17%	6.70%
SimCLR + k-NN (k=5)	3.58%	5.62%	3.57%	3.26%

The supervised FoodNet outperforms SSL linear probing by a large margin, since SL directly optimises the 251-class classification objective with label information. The SSL approach demonstrates that meaningful representations can be learned without any labels; however, the accuracy gap underscores the challenge of applying contrastive pre-training on a fine-grained, small-scale sub-sampled dataset.

8 Limitations and Future Directions

8.1 Limitations

- **Sub-sampling:** Capping each class at 300 images (74,990 total vs. 118,475 available) reduces the diversity of training examples and limits achievable accuracy.
- **Input resolution:** Using 128×128 images discards fine-grained texture details that may be discriminative for certain food categories.
- **SSL pre-training scale:** Contrastive learning benefits from large batch sizes and many epochs. GPU quota constraints limited SimCLR to 30 epochs with batch size 128.
- **Noisy labels:** The training set contains web-crawled images with label noise, which can hurt both SL and SSL performance.

8.2 Future Directions

- Train on the full 118k-image dataset using more powerful hardware or distributed training.
- Explore larger input resolutions (224×224) and pre-trained backbone feature extractors via transfer learning.
- Apply more advanced SSL methods such as DINO (6) or MoCo v3.
- Investigate semi-supervised learning to leverage both labelled and unlabelled data.
- Apply focal loss or class-balanced sampling to address label imbalance.

Statement of Academic Integrity

We, the undersigned, declare that this report and the accompanying software code represent our own original work. We have not used any form of plagiarism, including the use of large language models (such as ChatGPT or similar AI tools) to generate text, code, or any other part of the submitted material. All external sources, papers, and libraries used are properly cited throughout this document.

References

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A Figures

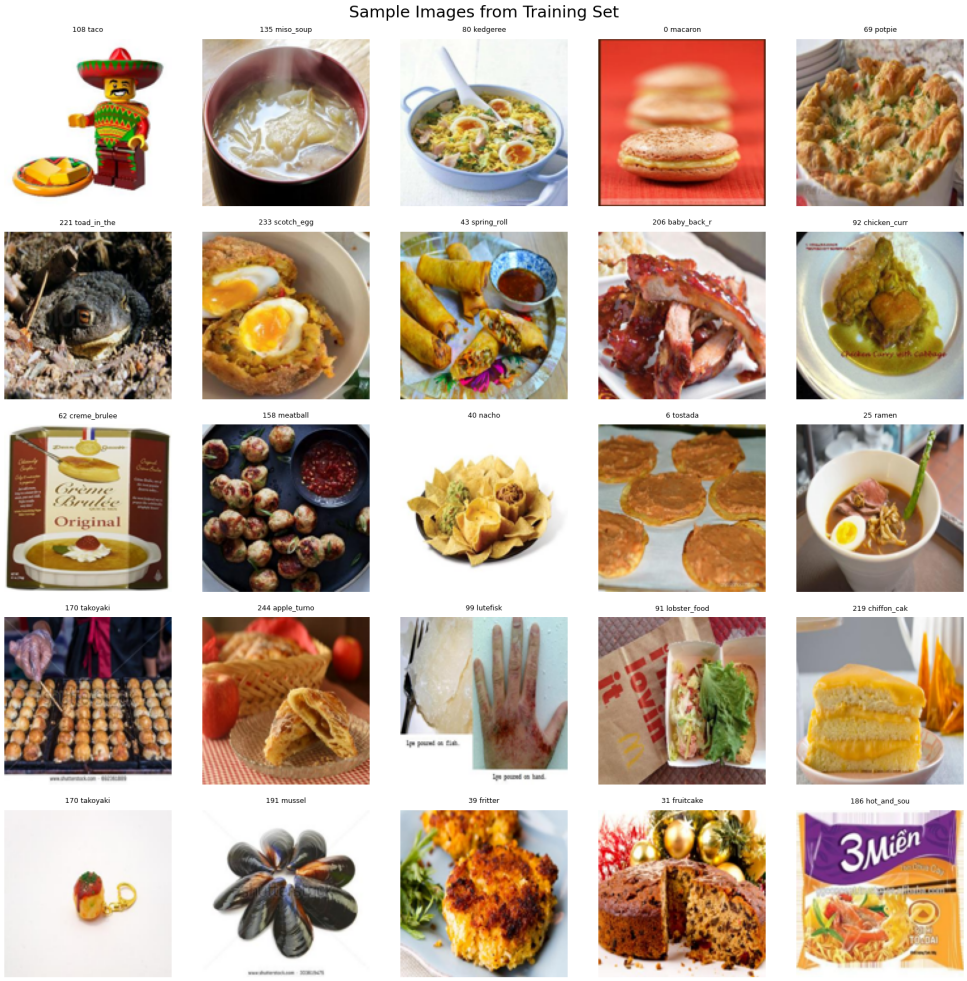


Figure 1: Sample images from the FoodX-251 training set. The dataset contains significant cross-domain noise (e.g., a LEGO figurine labelled as "taco", a hand image labelled as "lutefisk"), illustrating the challenges of web-crawled training data.

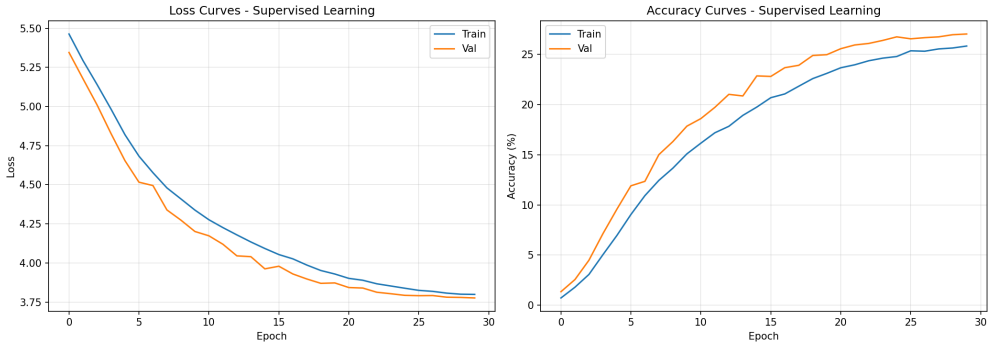


Figure 2: Training and validation loss (left) and accuracy (right) over 30 epochs for FoodNet under supervised learning. Both curves converge smoothly with cosine annealing, reaching a final validation accuracy of 27.04%. The validation accuracy consistently leads the training accuracy, indicating effective regularisation via label smoothing and dropout.

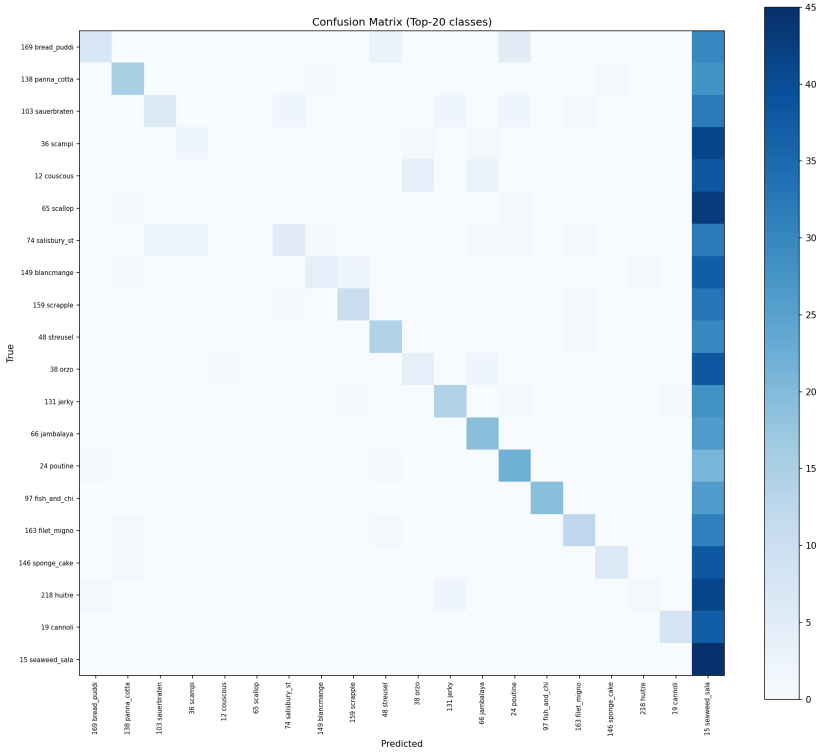


Figure 3: Confusion matrix for the 20 most frequent validation classes. The diagonal entries (correct predictions) are most prominent for visually distinctive categories such as *panna_cotta*, *jambalaya*, and *poutine*. The last column shows many samples being misclassified as *seaweed_sala*, suggesting bias towards this high-frequency class.

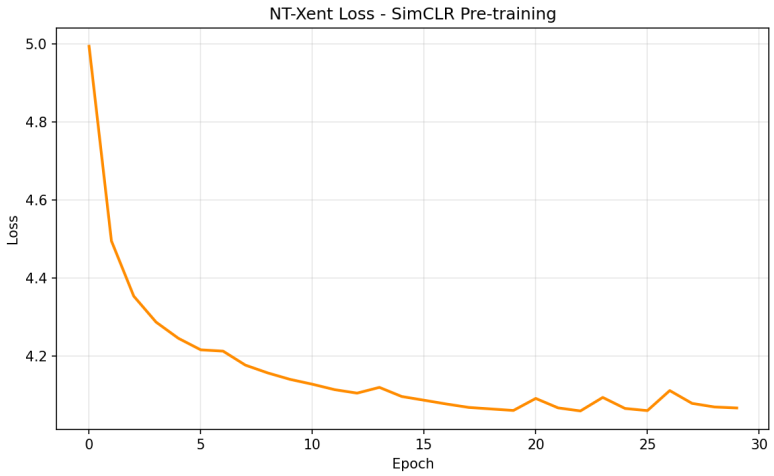


Figure 4: NT-Xent contrastive loss during SimCLR pre-training over 30 epochs. The loss decreases rapidly in the first 5 epochs (from 4.99 to 4.22) and then converges more gradually to 4.06, confirming that the backbone progressively learns more discriminative representations.

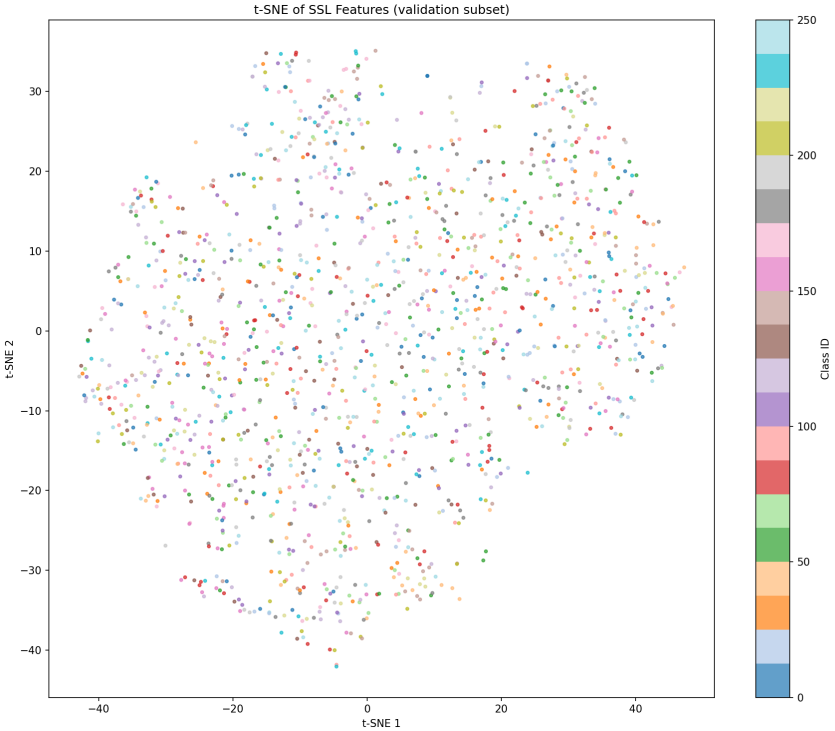


Figure 5: t-SNE projection of 256-dimensional SSL features extracted from the validation subset (1,500 samples). The absence of clear class clusters confirms that 30 epochs of SimCLR pre-training on a sub-sampled dataset are insufficient to produce well-separated representations for all 251 fine-grained food categories.

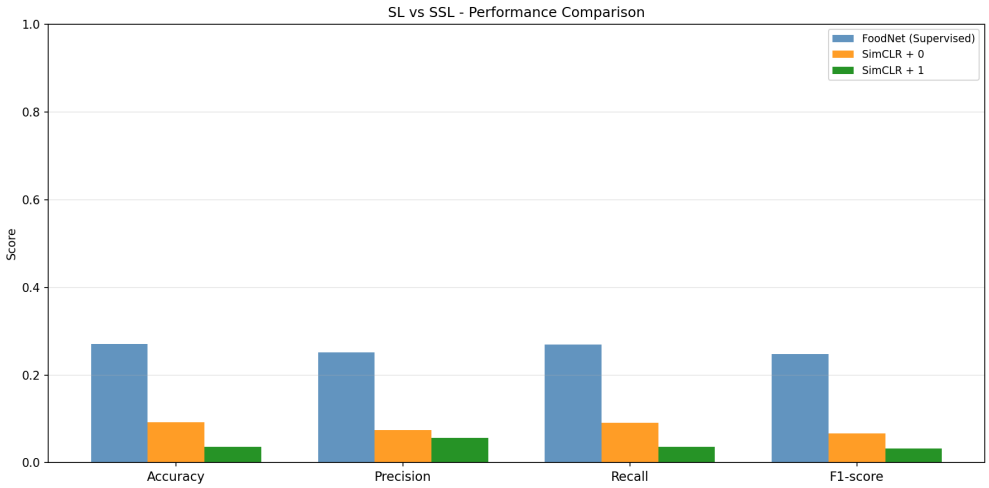


Figure 6: Performance comparison across all three methods on the 251-class validation set (macro-averaged metrics). FoodNet (Supervised) achieves 27.04% accuracy, substantially outperforming SimCLR + SVM (9.20%) and SimCLR + k-NN (3.58%). The legend labels "SimCLR + 0" and "SimCLR + 1" refer to Linear SVM and k-NN respectively.